
Privacy Analysis of Augmented Reality Glasses

12.15.2024 ~ 12.23.2025

Weekly Report 13

Michelle Jessica Bösigler
michellej.boesiger@unibas.ch

Department of Mathematics and Computer Science
University of Basel

23 December 2024

What has been done (Previous Week)

- Decryption efforts for Network traffic:
 - I received another rooted phone that already has mitmproxy as a system installed certificate but that also wasn't able to capture decrypted traffic. The problem I encountered on this phone is during the start of the Frida server.
 - While trying to obtain more information about the network traffic, I found and contacted a person who was in cooperation with Meta security in regards to the network safety of the glasses. His contribution was mainly towards the older version, the Rayban stories. But he was able to find a vulnerability which has since been fixed, and this vulnerability he found is exactly the direction we were attempting to attack from.
 - Due to his help I was able to get an insight on some technical aspects of the glasses.
- Wrote and uploaded documentations on the step by steps on what has been attempted the past weeks in regards to decrypting both Network (Bypass Certificate Pinning) and Bluetooth traffic. This includes the challenges faced and methods tested so far.
- Finished the documentation for the initial challenges, especially in regards to TWRP, when rooting the phone.
- Reorganized and uploaded the code for the visualizations, though now with a better structure of what images I generate, there is much more that I know how to compact.
- Reviewed the four literature provided and noted what I am and could be able to reference when doing the analysis. One of the paper is extremely similar to what I am doing / have done so far and is also very helpful in providing a structure for my thesis. It also gave me an inspiration how how do I define encrypted/decrypted. Though I hope to be able to talk through all this with Nima before the finalizing it.
- Begun writing the thesis. While I don't have the official formatting yet, this won't affect the drafting process at this stage. So far I have only written the introduction and started on the methodology.

What will be done next week

- I went through the experiments done and plan to redo some of the captures for a more refined analysis, since I also wanted to test it out with the new phone.
- Continue writing the bachelor thesis, most likely expanding in the sections, Experiment and Methodology.

- As mentioned, will update the code and generate some more images for the upcoming captures(Saturday/Sunday).
- Look more into other literature that could be related.
- Will continue looking into the decryption though that will not be the main focus (this is from my judgement with the time I have left, not sure if this is the correct step).

Meeting

Meeting canceled.